

4. Peaks of LH and FSH production occur during
 - (A) the menstrual flow phase of the uterine cycle.
 - (B) the beginning of the follicular phase of the ovarian cycle.
 - (C) the period just before ovulation.
 - (D) the secretory phase of the uterine cycle.
5. During human gestation, rudiments of all organs develop
 - (A) in the first trimester.
 - (B) in the second trimester.
 - (C) in the third trimester.
 - (D) during the blastocyst stage.

Levels 3-4: Applying/Analyzing

6. Which of the following is a true statement?
 - (A) All mammals have menstrual cycles.
 - (B) The endometrial lining is shed in menstrual cycles but reabsorbed in estrous cycles.
 - (C) Estrous cycles are more frequent than menstrual cycles.
 - (D) Ovulation occurs before the endometrium thickens in estrous cycles.
7. For which of the following is the number the same in human males and females?
 - (A) interruptions in meiotic divisions
 - (B) functional gametes produced by meiosis
 - (C) meiotic divisions required to produce each gamete
 - (D) different cell types produced by meiosis
8. Which statement about human reproduction is true?
 - (A) Fertilization occurs in the vagina.
 - (B) Spermatogenesis and oogenesis both require normal body temperature.
 - (C) An oocyte completes meiosis after a sperm penetrates it.
 - (D) The earliest stages of spermatogenesis occur closest to the lumen of the seminiferous tubules.

Levels 5-6: Evaluating/Creating

9. **DRAW IT** In human spermatogenesis, mitosis of a stem cell gives rise to one cell that remains a stem cell and one cell that becomes a spermatogonium. (a) Draw four rounds of mitosis for a stem cell, and label the daughter cells. (b) For one spermatogonium, draw the cells it would produce from one round of mitosis followed by meiosis. Label the cells, and label mitosis and meiosis. (c) Explain what would happen if stem cells divided like spermatogonia.

10. **EVOLUTION CONNECTION** Hermaphroditism is often found in animals that are fixed to a surface. Motile species are less often hermaphroditic. Explain why.
11. **SCIENTIFIC INQUIRY** Suppose that you discover a new egg-laying worm species. You dissect four adults and find both oocytes and sperm in each. Cells outside the gonad contain five chromosome pairs. Lacking genetic variants, explain how you would determine whether the worms can self-fertilize.
12. **WRITE ABOUT A THEME: ENERGY AND MATTER** In a short essay (100–150 words), discuss how different types of energy investment by females contribute to the reproductive success of a frog, a chicken, and a human.
13. **SYNTHESIZE YOUR KNOWLEDGE**



A female Komodo dragon (*Varanus komodoensis*) kept in isolation in a zoo had offspring. Each of the offspring had two identical copies of every gene in its genome. However, the offspring were not identical to one another. Based on your understanding of parthenogenesis and meiosis, propose a hypothesis to explain these observations.

For selected answers, see Appendix A.